FEASIBILITY STUDY

Sheikh Najjar Solar PV Project

Feasibility study for a 60 MWp solar plant with optional battery storage supplying Sheikh Najjar Industrial City, Aleppo

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Portfolio case study. The project and its sponsor, Halab Solar Development Company, are fictional. Market, resource, price and regulatory data are real and dated to their sources; all other inputs are labelled assumptions.

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1. Executive summary and recommendation

Halab Solar Development Company (the Sponsor, a fictional special-purpose developer used for this study) proposes to build, own and operate a 60 MWp / 50 MWac ground-mounted solar photovoltaic plant at Sheikh Najjar Industrial City, north-east of Aleppo, and to sell its output under long-term power purchase agreements (PPAs) to manufacturers inside the city. Sheikh Najjar is Syria's largest industrial estate, covering about 4,412 hectares with 6,048 industrial plots [36]; its management reported 1,553 producing facilities in July 2026, with a further 704 under construction and 350 being rehabilitated [34]. Electricity remains the binding constraint: reporting in 2026 describes an industrial city that needs more power than it receives, with factories running part-time or on diesel generators [35].

The economic case rests on a wide price gap. Rationing-exempt industrial users pay 17 Syrian pounds (SYP) per kWh under Decision 687, effective 1 November 2025 [11], equivalent to 13.9 US cents at the official rate of 122.5 SYP/USD on 13 October 2026 [15]. Self-generation with industrial-grade diesel at 150 SYP per litre [13] costs about 37.3 cents per kWh once fuel and generator running costs are included. A USD-indexed daytime PPA at 10 cents per kWh would be 28% below the grid tariff and 73% below diesel, while still covering the plant's levelised cost of 7.85 cents per kWh at the project's risk-adjusted cost of capital.

Table 1.1 Key results, base case

Indicator	Value
Installed capacity	60 MWp DC / 50 MWac, fixed tilt 31 degrees
Year-1 energy delivered	96.8 GWh (1,613 kWh/kWp; AC capacity factor 22.1%)
Total capital cost	USD 32.2 million (USD 537/kWp; USD 645/kWac)
Total funding incl. fees and interest during construction	USD 33.6 million: senior debt 20.2, equity 13.4
PPA tariff	USD 0.100/kWh in year 1, indexed 1% a year
Project NPV at WACC of 15.4%	USD 9.9 million
Project IRR (after tax, unlevered)	20.4%
Equity IRR / cost of equity	28.8% / 24.3%
Equity NPV at cost of equity	USD 2.8 million
Payback (project / equity), years from COD	4.8 / 3.6
LCOE at WACC / at 8%	USD 0.0785 / 0.0588 per kWh
Minimum / average DSCR (12-year loan)	1.91x / 2.36x

On the base case the project clears every financial test: a positive NPV of USD 9.9 million, a project IRR about 5 percentage points above the WACC, an equity IRR above a demanding 24.3% cost of equity, and debt cover with a minimum DSCR of 1.91x. The PPA tariff that sets project NPV to zero is USD 0.082/kWh, and the tariff that just meets the cost of equity is USD 0.092/kWh.

The base case is, however, a most-likely case, and the main risks lean to the downside. A 10,000-run Monte Carlo simulation over yield, capex, opex, currency conversion, collections, interest rate, degradation and availability gives a median project NPV of USD 2.6 million and a 30% probability of negative NPV when those risks are left unmitigated. When the simulation reflects the contractual protections recommended in this study (fixed-price EPC, a currency convertibility undertaking with political risk insurance, and payment security from offtakers), the probability of negative NPV falls to 1.3% and the median equity IRR rises to 27.2%. The investment case therefore depends on contract structure more than on technology.

Recommendation. The study recommends that the Sponsor proceed to the bankable feasibility and development phase, and that financial close be made conditional on the following:

1. Take-or-pay PPAs covering at least 80% of P50 output with creditworthy offtakers, at a year-1 tariff of at least USD 0.100/kWh (and never below USD 0.092/kWh), indexed in US dollars and invoiced in SYP at the Central Bank of Syria rate.
2. Written confirmation from the Ministry of Energy that direct sale to industrial users over the zone network is licensable under Electricity Law 32/2010 as amended by Law 41/2022, since Decision 154 of 2025 covers only 2-10 MW plants selling to the state.
3. A currency convertibility and transfer undertaking under the Investment Law, political risk insurance, and an offshore debt service reserve.
4. A fixed-price, date-certain EPC contract keeping total capex at or below about USD 35.8 million (USD 596/kWp), the level at which equity still earns its cost.
5. At least six months of on-site irradiance measurement confirming the PVGIS-based P50 within about 5%.

The battery option (20 MW / 80 MWh) should not be included in the first phase. At an evening tariff of USD 0.16/kWh it reduces project NPV by USD 9.1 million and earns an incremental IRR of only 1.0%; it breaks even only at an evening tariff of about USD 0.22/kWh. That is still below the cost of diesel, so a battery could be added once offtakers sign evening contracts at that level. Single-axis trackers raise modelled yield by 22% and NPV by USD 6.0 million under indicative cost assumptions; the EPC tender should therefore price both fixed-tilt and tracker designs.

2. Project description

2.1 Overview and objectives

The project is a utility-scale solar plant sized to supply daytime electricity to manufacturers in Sheikh Najjar Industrial City. It has three aims: to give industrial users a cheaper and more reliable daytime supply than grid power and diesel generators; to earn a return for investors that reflects Syrian country risk; and to show that private, contracted renewable generation can be financed inside an industrial zone. The plant would sit on about 78 hectares of industrial-city land, less than 2% of the city's area [36], and connect to the city's 20 kV distribution network through a new step-up substation.

Table 2.1 Project parameters

Parameter	Base case	Basis
Location	Sheikh Najjar Industrial City, Aleppo (indicative 36.27 N, 37.30 E; elevation about 540 m)	Assumption; PVGIS elevation [1]
DC / AC capacity	60 MWp / 50 MWac (DC/AC ratio 1.20)	Assumption
Modules	Bifacial n-type TOPCon, about 620-650 Wp class	Market standard [5]
Mounting	Fixed tilt, 31 degrees, south-facing (tracker option evaluated)	PVGIS optimum [1]
Inverters	String inverters, 300 kW class	[8]
Land	About 78 ha at 1.3 ha/MWp	Assumption; 2021 plant 33 MW on 35 ha [32]
Grid connection	Step-up substation into the city 20 kV network	Assumption
Offtake	Bilateral take-or-pay PPAs with 15-30 industrial users	Assumption
Construction / COD	12 months; COD September 2028	Assumption
Operating life modelled	25 years	Assumption
Optional storage	20 MW / 80 MWh lithium iron phosphate battery	Assumption

2.2 Sponsor and contractual structure

The Sponsor would hold the plant through a Syrian private joint-stock company, registered with an object clause covering generation and sale of renewable electricity; Syrian legal commentary recommends this form for plants above 10 MW [40]. Senior debt would come from development finance institutions and regional banks with political risk cover. The SPV would sign a land lease with the industrial city administration, a network use and metering agreement with the Syrian Electricity Company, an engineering, procurement and construction (EPC) contract, an operations and maintenance (O&M) contract, and PPAs with individual factories. Section 5 discusses the licensing route; Section 7 covers how the SPV would be run.

2.3 Relationship to existing solar capacity at the site

Sheikh Najjar already hosts Syria's first large solar plant. Its foundation stone was laid in April 2021 for 33 MW under a contract worth EUR 36 million, with 86,840 modules on 350,000 square metres and expected output of 52 GWh a year [32]. Global Energy Monitor lists a 42 MW first phase operating since 2021 and a 58 MW second phase under construction [33]; these sources differ and the operating capacity should be confirmed during development. In July 2026 the industrial city also tendered space above and around its water tanks for

photovoltaic generation [39]. The proposed plant would add to this capacity rather than replace it, and its PPA model differs: the existing plant feeds the public network, whereas this project would contract directly with factories.

3. Market and demand analysis

3.1 National power context

Syria's installed capacity of about 8.5 GW in 2011 has fallen to roughly 2 GW of available capacity, leaving many regions with two to four hours of supply a day and creating heavy reliance on diesel generation [18]. The government's near-term target is about eight hours a day nationally, supported by gas imports through Türkiye that first supplied the Aleppo thermal plant [18]. Supply in Aleppo remains volatile: some neighbourhoods reached about 16 hours a day at the end of August 2026, but a maintenance outage at the thermal plant and a fault on the Aleppo 230 kV line from 16 September 2026 pushed rationing above 15 hours a day in some areas [16]. The Syrian Electricity Company, created by Decree 45 of April 2026, announced further temporary rationing on 17 September 2026 after four generating units went offline [17].

Tariffs have moved towards cost recovery. Decisions 686 and 687 raised the price for rationing-exempt industrial, commercial and tourism users to 1,700 old SYP (17 new SYP) per kWh and for smelting, rolling and other heavy users to 1,800 old SYP, from 1 November 2025 [11]. Industrialists told Enab Baladi that the increase from 1,500 SYP came as a shock [11]. A 2026 IDOS study puts the cost of producing grid electricity at about USD 0.12-0.18 per kWh [19], which suggests tariffs are unlikely to fall in real terms.

3.2 Sheikh Najjar Industrial City

Table 3.1 Sheikh Najjar Industrial City: reported indicators

Indicator	Value	Date	Source
Total area	about 4,412 ha	2026	[36]
Industrial plots	6,048 (textiles 1,825; engineering 1,465; chemicals 1,389; food 1,043)	2026	[36]
Producing facilities	960	Mar 2025	[38]
Producing facilities	1,506 (about 550 at full capacity, 200 partial)	end-2025	[35]
Producing facilities	1,553; 704 under construction; 350 in rehabilitation	Jul 2026	[34]
New factories entering production	546 in 2025	2026	[36]
Direct and indirect jobs	over 80,000	Jul 2026	[34]
Recent substation additions	60 MW and 30 MW	Jan 2023	[37]

The city has grown quickly since the end of 2024, but its electrical network was described by its own management as dilapidated in March 2025 [38], and in 2026 it planned more than 35 infrastructure projects [36]. This combination of rising demand and weak supply is the market opening for a dedicated plant.

3.3 Industrial load estimate

No published figure exists for the city's current electricity demand; this is a data gap. The study therefore estimates daytime demand bottom-up from the 1,553 producing facilities [34] and an assumed average coincident daytime load per facility. The textile, chemical and engineering mix suggests a wide range, so three cases are shown. Even the low case exceeds the plant's 50 MWac output, so the plant can be fully absorbed by daytime load without exporting to the national grid, provided enough factories sign PPAs.

Table 3.2 Estimated coincident daytime demand (assumption-based)

Case	Average load per facility	Zone daytime demand	Plant share (50 MWac)
Low	60 kW	93 MW	54%
Base	100 kW	155 MW	32%
High	150 kW	233 MW	21%

Average loads are analyst assumptions, not measured data. A metering survey of candidate offtakers is a development priority.

3.4 Price comparators and willingness to pay

Figure 3.1 compares the proposed PPA tariff with the alternatives open to a factory. Diesel costs are built up from the September 2026 prices: standard diesel at 175 SYP/L from 13 September [12] and the 150 SYP/L grade for productive sectors announced on 17 September, when the Energy Minister also put the actual cost of diesel at 206 SYP/L [13]. At the official rate this is USD 1.22 and 1.43 per litre; GlobalPetrolPrices reported USD 1.496 per litre for standard diesel on 5 October 2026 [14]. Generator fuel use is assumed at 0.28 L/kWh, above the 202 g/kWh (about 0.24 L/kWh) a large modern set achieves at full load [30], because factory generators usually run at part load. Adding an assumed USD 0.03/kWh for maintenance and capital recovery gives 37.3 cents per kWh with industrial diesel.

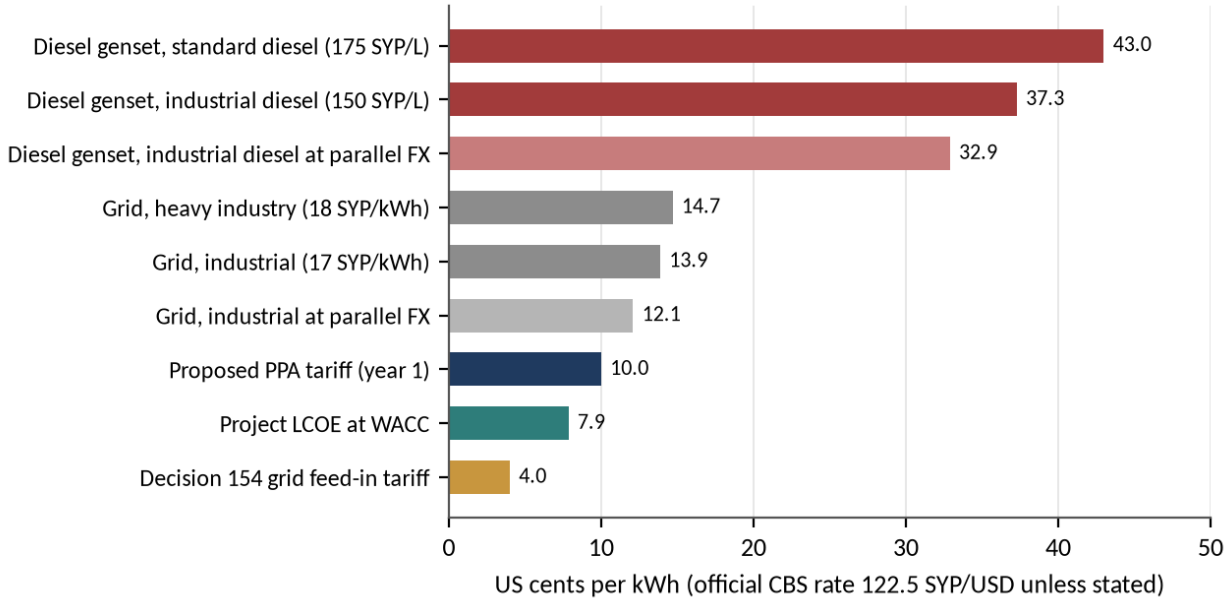


Figure 3.1 Cost of electricity options for a Sheikh Najjar factory (US cents per kWh)

Sources: tariffs [11]; diesel [12, 13]; FX [15]; LCOE from the financial model. Diesel includes an assumed 3 cents/kWh non-fuel cost.

Because grid supply is intermittent, a factory's real cost of power is a blend of grid and diesel. Figure 3.2 shows the blended cost for a plant working 16 hours a day: at eight hours of grid supply it is about 25.6 cents per kWh, and it stays above the PPA tariff even with full grid supply. The PPA therefore saves money in every supply state, which supports willingness to pay. The study does not treat willingness to pay as the binding constraint; the constraints are offtaker credit quality, enforceability of contracts and currency convertibility.

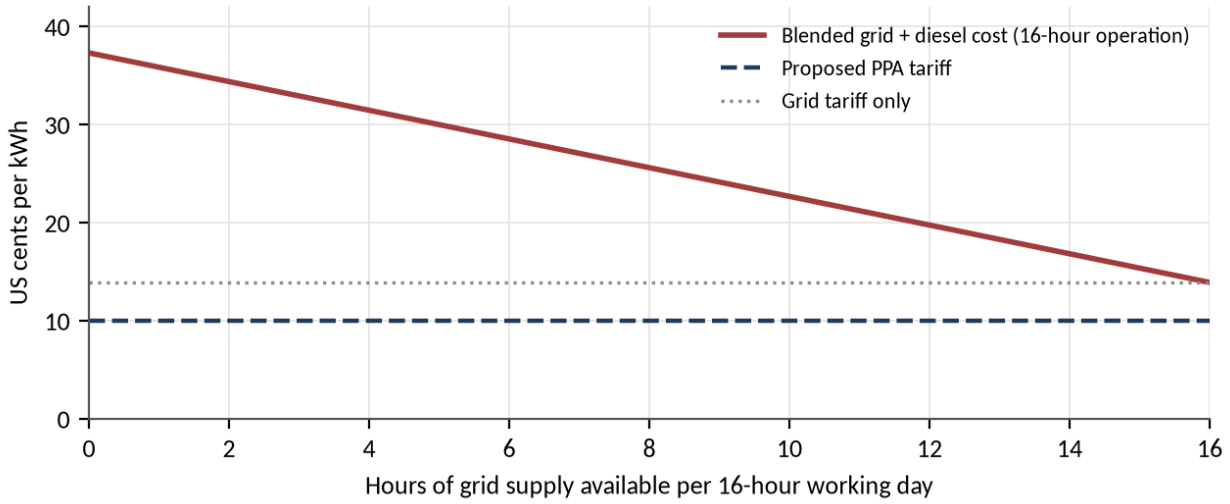


Figure 3.2 Blended grid and diesel cost by hours of grid supply

Source: model calculation using the grid and diesel costs in Figure 3.1.

3.5 Competition and market risks

Three developments could erode the price advantage. First, grid supply may improve: the UCC-led consortium signed concessions in November 2025 for 4,000 MW of gas-fired plants, including 1,200 MW in North Aleppo, plus 1,000 MW of solar of which 300 MW is in Aleppo [21], and gas plants are expected within about three years [21, 22]. Second, the grid tariff is fixed in SYP, so its US dollar value falls if the pound weakens; at the 13 October 2026 parallel rate of 140.5 SYP/USD it is already only 12.1 cents [15]. Third, factories can install their own rooftop solar, as more than 2,200 MW of off-grid systems nationally show [19]. These risks favour long take-or-pay contracts signed early, a tariff with a clear discount to the grid, and a reliability offer that rooftop systems cannot match at scale.

4. Technical feasibility

4.1 Site

The indicative site lies inside or immediately next to the industrial city, about 4 km east of the coordinates recorded for the existing plant (36.2701 N, 37.2569 E) [33], at about 540 m elevation [1]; terrain, ground conditions and drainage must be confirmed by topographic and geotechnical survey. NASA POWER gives a mean air temperature of 17.4 °C, rising to 30.2 °C in July [3], so heat losses are material but moderate. The final plot must be agreed with the city administration and checked for unexploded ordnance, ground conditions and drainage, which are standard development items in a former conflict area.

4.2 Solar resource

Table 4.1 Solar resource at the site from three independent datasets

Dataset	Period	GHI (kWh/m²/yr)	In-plane (kWh/m²/yr)	PV output (kWh/kWp/yr)
PVGIS 5.3, SARAH3 [1]	2005-2023	n/a	2,168 (31 degree tilt)	1,654.5 (14% system loss)
Global Solar Atlas (Solargis) [4]	long-term to 2025	1,942	2,212 (optimum)	1,761
NASA POWER [3]	2001-2025	1,864	n/a	n/a

The datasets agree within about 4% on global horizontal irradiance. The model uses the PVGIS output of 1,654.5 kWh/kWp as P50 because it is the lower of the two yield estimates and is reproducible; the Global Solar Atlas figure is 6.5% higher and is treated as upside. Monthly yield ranges from about 92 kWh/kWp in January and December to 173 kWh/kWp in July and August (Figure 4.1).

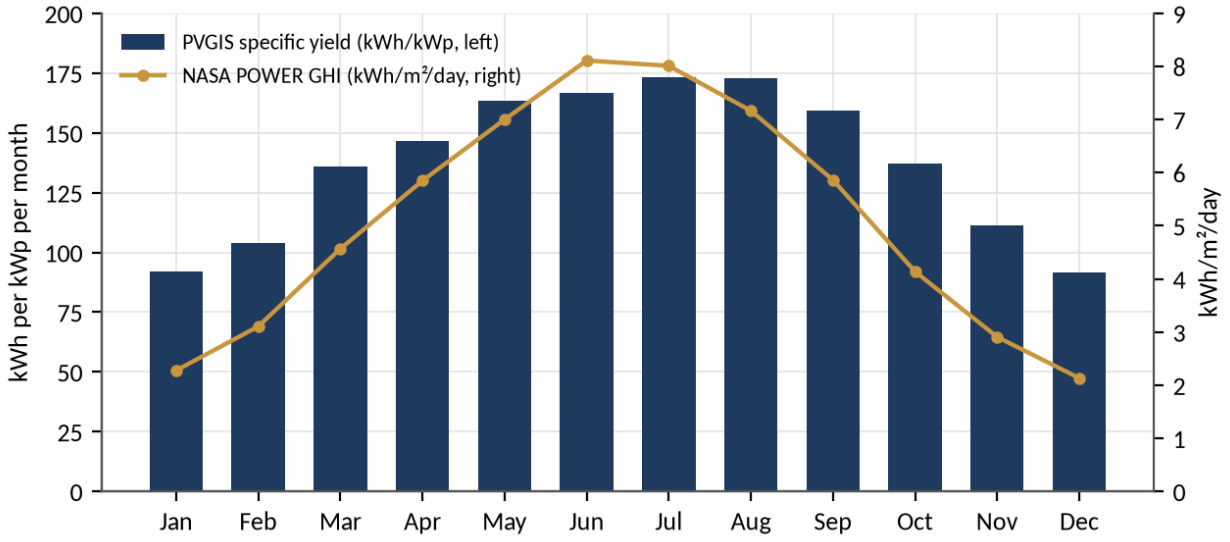


Figure 4.1 Monthly PV yield (PVGIS) and global horizontal irradiance (NASA POWER)

Sources: [1, 3].

4.3 Energy yield

PVGIS reports total losses of 23.7% between in-plane irradiance and AC output, of which 8.4% comes from temperature and low irradiance and the rest from reflection, spectrum and a 14% allowance for soiling, wiring, mismatch and inverter losses [1]. The model then applies 98.5% availability for the plant and the offtakers' network, and an inverter clipping loss of 0.03% calculated from the 2019-2023 hourly series at a DC/AC ratio of 1.20 [2]. Year-1 delivered energy is 96.8 GWh, a delivered specific yield of 1,613 kWh/kWp.

For debt sizing the study uses a P90 estimate. Combining the PVGIS year-to-year standard deviation of 41 kWh/kWp (2.5%) with an assumed 5% modelling uncertainty gives a total of 5.6%, and P90 is 92.8% of P50. Debt service is covered at P90 with a minimum DSCR of 1.74x.

4.4 Degradation

Modules are assumed to lose output in line with a current TOPCon manufacturer warranty: at most 1.0% in the first year and 0.4% a year thereafter, leaving at least 87.4% after 30 years [9]. Using warranted rather than typical degradation is conservative. Delivered energy falls from 96.8 GWh in year 1 to 87.4 GWh in year 25 (Figure 4.2).

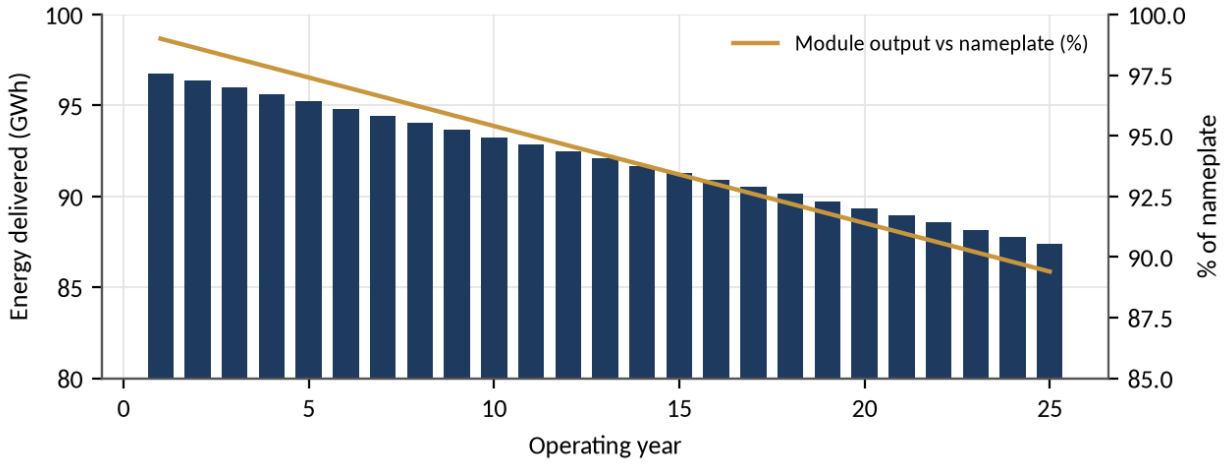


Figure 4.2 Annual energy delivered and module output factor

Source: model, degradation per [9].

4.5 Capacity sizing and design choices

Three considerations set the size. First, the plant must stay below daytime demand of contracted offtakers so that no energy is spilled: at 50 MWac it represents about a third of the base-case daytime demand estimate. Second, it must be large enough to spread fixed costs (substation, security, management) and attract institutional lenders; projects below about 10 MW would also fall under Decision 154 and its USD 0.04/kWh tariff [20]. Third, it must be deliverable within about a year. The hourly profiles in Figure 4.3 show that average hourly output peaks at about 44-45 MW from June to September, so the 50 MWac limit clips very little energy.

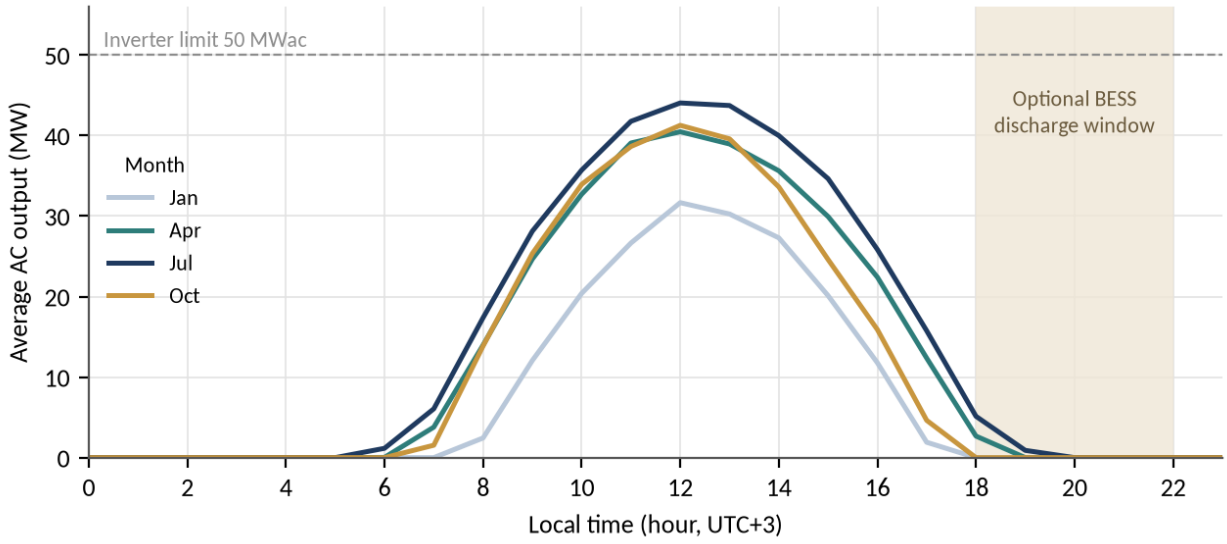


Figure 4.3 Average hourly AC output by month, 60 MWp plant (local time)

Source: PVGIS hourly series 2019-2023 [2], scaled to plant size.

Table 4.2 Mounting options (indicative)

Indicator	Fixed tilt (base)	Single-axis tracker
Specific yield, PVGIS (kWh/kWp)	1,655	2,020
Capex (USD million)	32.2	36.6
LCOE at WACC (USD/kWh)	0.0785	0.0731
Project NPV (USD million)	9.9	16.0
Equity IRR	28.8%	33.1%
O&M and spares	Simple, no moving parts	Motors and controllers; specialist spares

Tracker case assumes USD 0.06/Wp extra structure cost, USD 2/kWp-yr extra O&M and 1.6 ha/MWp; these are assumptions to be tested in the EPC tender.

Fixed tilt is kept as the base case because it is simpler to operate and maintain where specialist spares are hard to obtain and security is a concern. The tracker result is strong enough that both options should be priced.

4.6 Battery storage option

A 20 MW / 80 MWh battery would charge from the plant between 09:00 and 17:00 and discharge from 18:00 to 22:00 to serve evening shifts. An hourly simulation over 2019-2023 shows it would absorb 30.5 GWh a year and deliver 26.5 GWh after an assumed 87% round-trip efficiency, about 331 full cycles a year, reaching full charge on 93% of days. Turnkey system prices fell to USD 117/kWh globally in 2025 [7], and Ember estimates all-in costs of USD 125/kWh outside China and the US [7]; the study assumes USD 165/kWh to allow for logistics and country risk. The technical case is straightforward; Section 8.9 shows that the commercial case is not yet there.

5. Legal and regulatory review

5.1 Sector law and licensing

Electricity Law No. 32 of 2010 remains the core statute. It places policy, licensing and tariff-setting with the Ministry of Energy, allows private generation and lets private and industrial producers connect to the grid and sell surplus power [18]. Laws No. 32 of 2021 and No. 41 of 2022 amended it so that the tariff in a PPA with the national utility can be agreed contractually and approved by the Council of Ministers, and Law 41 of 2022 clarified that licensed investors may in some circumstances sell electricity over private networks [18]. Generation projects need a Ministry licence and Council of Ministers approval of the investment framework [18].

Decision No. 154 of 3 July 2025 created a fast-track licence for 2-10 MW renewable plants: a 20-year renewable licence and state purchase at USD 0.04/kWh for solar, USD 0.06/kWh for solar with four-hour storage, paid monthly in SYP at the official rate [20]. The IDOS study notes that Decision 154 treats the state as the sole buyer, while the 2021 and 2022 amendments allow peer-to-peer sales at higher prices of about USD 0.08/kWh [19]. This project is above the Decision 154 size range and relies on direct sale, so its licence must be negotiated individually. That is the single most important legal condition, and Section 9 shows the project does not work at the Decision 154 tariff.

Institutions are changing. The Ministries of Electricity, Oil and Water were merged into a Ministry of Energy [19], the Syrian Electricity Company was formed as a state holding company by Decree 45 of 2026 [17], and the Ministry has said it plans an independent electricity regulator with rules for feed-in tariffs, wheeling and peer-to-peer trading [19]. A wheeling or network-use charge is not yet published, so the model assumes USD 0.005/kWh.

5.2 Investment incentives and tax

Investment Law No. 18 of 2021, amended by Decree 114 of 2025, offers income-tax reductions of 50-75% for ten years for industrial and energy projects, full customs exemption on imported equipment, 100% foreign ownership, profit repatriation after tax and access to international arbitration [25]. The investment law also protects against expropriation except for public use with compensation in convertible currency and contains stability provisions [18]. A draft income-tax law published for consultation in 2025 would set corporate tax at 10% for priority sectors and 15% for others [26]; this study found no confirmation that it has been enacted and models 10% with a 75% reduction for ten years. The tornado analysis shows tax has little effect on returns.

5.3 Sanctions and financial access

The sanctions picture has changed since December 2024. The United States removed its Syria sanctions programme from 1 July 2025, the EU lifted economic measures except those on security grounds in May 2025, and the UK eased its regime in April 2025 [18]. The Caesar Act was repealed on 18 December 2025 and Syrian banks were reconnected to SWIFT on 20 November 2025 [25]. Designations of named individuals and entities and export controls still apply, so all counterparties need sanctions and anti-money-laundering screening [18].

5.4 Recent transactions

Table 5.1 Recent solar and power transactions in Syria

Date	Transaction	Relevance	Source
Jun 2025	World Bank USD 146 million grant (Syria Electricity Emergency Project) for transmission rehabilitation	Multilateral re-engagement	[24]
Jul 2025	Decision 154; MoU with 20Solar Energy for two 100 MW plants, one with storage	Tariff benchmark; storage interest	[20]
Nov 2025	Final concessions with UCC-led consortium: 4,000 MW gas, 1,000 MW solar incl. 300 MW in Aleppo	Competing supply in Aleppo	[21]
Jul 2026	Aleppo industrial city tender for PV over water tanks	In-zone procurement precedent	[39]
Aug 2026	Syrian Electricity Company and Al-Harfi: 760 MW solar and 1,077 MWh storage, Damascus countryside	Storage at scale; state offtake	[23]

5.5 Permits

Environmental impact assessment may be required under Ministerial Order No. 225 of 2008, overseen by the Ministry of Local Administration and Environment, and land-use permits are obtained at governorate and municipal level under Local Administration Law No. 107 of 2011 [18]. Inside an industrial city, land allocation runs through the city administration, which operates a one-stop window for investors [34].

6. Environmental and social impact

The main environmental benefit is displaced diesel. If every kWh from the plant replaced generator output, the plant would avoid about 27 million litres of diesel and 73 thousand tonnes of CO2 in year 1, using 0.28 L/kWh and the EPA factor of 10.21 kg CO2 per gallon (2.70 kg per litre) [31]. This is an upper bound: part of the energy will displace grid power, whose emission factor is not published for Syria (a data gap). Lower generator use also reduces local air pollution and noise in the city.

Table 6.1 Principal impacts and mitigation

Impact	Phase	Significance	Mitigation
Land take (about 78 ha)	Construction	Low: zoned industrial land	Use allocated plots; avoid productive farmland
Water use for cleaning in a water-stressed city	Operation	Medium	Robotic dry cleaning; rainwater storage
Dust, traffic and noise	Construction	Medium, temporary	Traffic plan; dust suppression; daytime working
Worker health and safety	Both	High if unmanaged	HSE plan aligned with IFC Performance Standard 2
Security personnel and community relations	Both	Medium	Code of conduct for guards (IFC PS4); grievance mechanism
Explosive remnants of war	Construction	Medium	Survey and clearance before mobilisation
End-of-life modules and batteries	Decommissioning	Medium	Take-back clauses; recycling plan
Electricity affordability	Operation	Positive	Tariff below grid and diesel; transparent indexation

Social sensitivity around energy prices is high. The November 2025 tariff increase triggered protests in January and February 2026 [19], and a 40% diesel price rise in September 2026 led to the widest protests since the change of government [13]. A PPA priced below the grid tariff, with a simple, published indexation formula, reduces this risk. Construction would employ an estimated 300-400 workers at peak and operation about 20-25 permanent staff (assumptions based on typical solar projects of this size), with a target of hiring locally and training technicians from Aleppo University and technical institutes.

7. Organisation and operations

The SPV would be run by a small management team, with technical work contracted out under performance-based agreements. A board with Sponsor and minority investor representatives would approve budgets, offtake policy and distributions; lenders would hold step-in rights and receive quarterly reports.

Table 7.1 Operating organisation

Function	Staffing / provider	Key responsibilities
General management	General manager, finance manager (SPV)	Budget, lender reporting, compliance, tax
Commercial	Offtake manager, two billing officers (SPV)	PPA management, metering, invoicing, collections, credit monitoring
Operations and maintenance	O&M contractor, about 12 technicians on site	Preventive and corrective maintenance, cleaning, spares, 99% availability target
Security	Licensed security contractor	Perimeter, CCTV monitoring, patrols, incident response
HSE and community	HSE officer (SPV)	ESMP implementation, grievance mechanism, training
Asset monitoring	SCADA with remote monitoring centre	Performance ratio, alarms, forecasting

Billing would use revenue-grade meters at each offtaker, read remotely, with monthly invoices in SYP at the Central Bank of Syria rate on the invoice date. Offtakers would post a bank guarantee or prepayment equal to three months of billing; if payment is late the SPV can suspend supply and re-allocate energy to other factories, which is easier inside one industrial city than across a national grid. The O&M contract would include an availability guarantee with liquidated damages, a two-year spare-parts stock and a training programme for local technicians.

8. Financial feasibility

8.1 Basis of preparation

The model is annual, in nominal US dollars, with capital spent in year 0 (construction, about 12 months to September 2028) and 25 operating years. Revenue is USD-indexed but received in SYP; currency risk is captured through an assumed 3% conversion loss in the base case and up to 15% in the sensitivity and Monte Carlo analysis. The Excel workbook delivered with this report contains every formula and was recalculated in LibreOffice and reconciled to the Python reference model (Appendix E).

8.2 Capital cost

Table 8.1 Capital cost estimate

Item	USD million	USD/kWp	Share
PV modules (TOPCon, bifacial)	6.84	114	21.2%
Freight, insurance, port and inland logistics	0.72	12	2.2%
Inverters	1.80	30	5.6%
Mounting structures	3.90	65	12.1%
Electrical balance of system	3.30	55	10.2%
Substation and interconnection	4.00	67	12.4%
Civil works and fencing	2.10	35	6.5%
Installation labour	2.40	40	7.4%
Security systems and SCADA	0.72	12	2.2%
EPC overhead and margin	2.06	34	6.4%
Development and permitting	1.20	20	3.7%
Owner's engineer and construction insurance	0.42	7	1.3%
Contingency	2.78	46	8.6%
Total capital cost	32.24	537	100.0%

Module price from InfoLink, 30 Sep 2026 [5]; other items are assumptions (Appendix A).

Modules are priced at USD 0.114/W, InfoLink's 30 September 2026 average for TOPCon modules sold outside China [5], plus USD 0.012/W for freight and inland transport. Inverter cost allows a margin over the USD 0.012-0.025/W bids in China's large 2024 tender [8]. The total of USD 537/kWp (USD 645/kWac) compares with IRENA's 2025 global weighted average for utility-scale PV of USD 667/kW [6], and is well below the 2021 Sheikh Najjar contract of EUR 36 million for 33 MW [32], which included eight years of deferred payment and was signed when module prices were considerably higher. A 10% contingency is included.

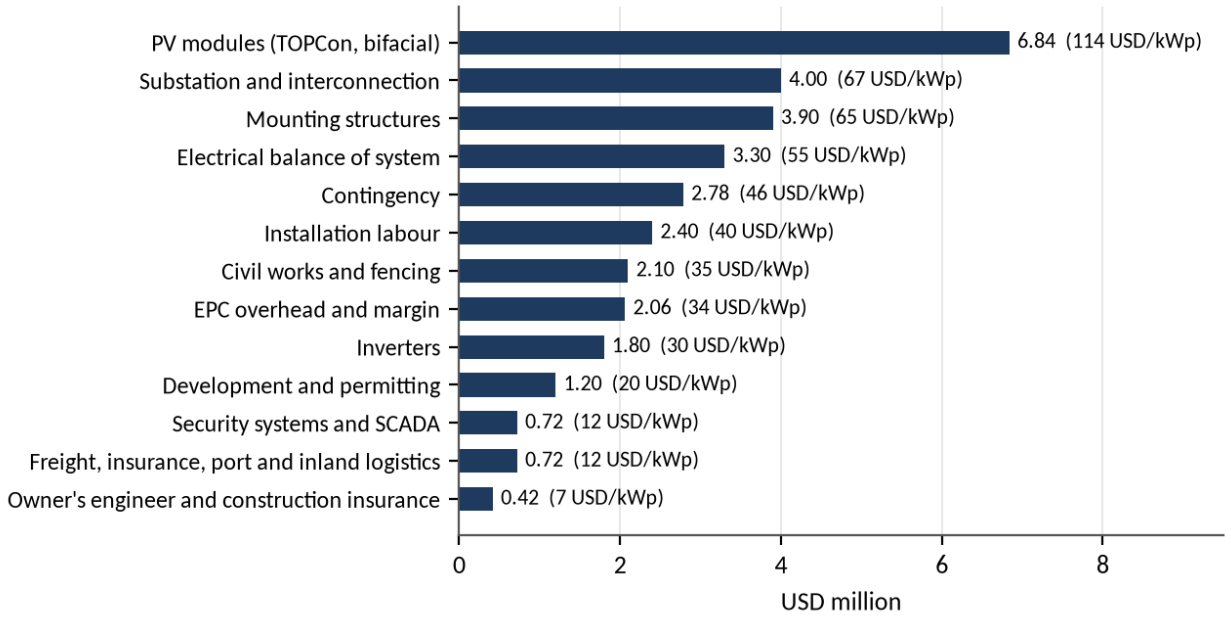


Figure 8.1 Capital cost breakdown

Source: model, Table 8.1.

8.3 Operating costs

Table 8.2 Year-1 operating costs

Item	USD	USD/kWp
O&M contract incl. cleaning	540,000	9.0
Property and business-interruption insurance	193,466	3.2
Political risk insurance	161,361	2.7
Security	250,000	4.2
Land lease	46,800	0.8
SPV administration, billing and metering	300,000	5.0
Network use and metering charge	483,870	8.1
Total year-1 opex	1,975,496	32.9

All items escalate at 2.5% a year. An inverter replacement of USD 1.5 million is assumed in year 13.

Year-1 opex of USD 33/kWp (about USD 40/kWac) is above the NREL benchmark of USD 22/kWac-yr for fixed O&M in the US [10] because it carries Syria-specific items: security, political risk insurance and network charges.

8.4 PPA tariff and revenue

The year-1 tariff of USD 0.100/kWh is set to give offtakers a clear saving against the grid (about 28%) while covering the project's cost of capital. It is indexed at 1% a year, below the 2.5% opex escalation, so margins narrow slightly over time. Year-1 revenue after an assumed 97% collection rate and the 3% conversion loss is USD 9.11 million, and EBITDA is USD 7.13 million, a margin of 78%.

8.5 Financing structure

Table 8.3 Sources and uses of funds

Uses	USD million	Sources	USD million
Capital cost	32.24	Senior debt (60%)	20.17
Upfront financing fees (2% of debt)	0.40	Sponsor equity (40%)	13.45
Interest during construction	0.97		
Total uses	33.62	Total sources	33.62

Table 8.4 Senior debt terms (assumed)

Term	Assumption	Basis
Lenders	DFI-led group with regional banks	Assumption
Interest rate	9.61% fixed: 7-year US Treasury 5.11% + 4.50% margin	Treasury yield 13 Oct 2026 [28]; margin assumed
Tenor and repayment	12 years from COD, equal annual instalments of USD 2.90 million	Assumption
Gearing	60% of total funding	Assumption
Covenants	Distribution lock-up at DSCR 1.15x; 1.30x reference minimum	Assumption
Security	Share pledge, assignment of PPAs and insurances, offshore accounts	Market practice

8.6 Discount rates

The cost of equity is estimated with the capital asset pricing model plus a country risk premium: the 10-year US Treasury yield of 5.21% on 13 October 2026 [28], plus a beta of 0.85 times Damodaran's mature-market equity risk premium of 4.23%, plus his Syria country risk premium of 15.54% (January 2026) [27]. This gives 24.35%. With a 9.61% cost of debt, 60% gearing and the 2.5% effective tax rate during the relief period, the after-tax WACC is 15.36%. SOFR stood at 3.87% on 9 October 2026 [29] and is shown for reference only, because the loan is assumed to be fixed-rate.

8.7 Projected cash flows

Table 8.5 Summary cash flows, selected years (USD million unless stated)

Year	Period	GWh	Revenue	Opex	EBITDA	Tax	CFADS	Debt service	Equity CF	DSCR
1	2028/29	96.8	9.11	1.98	7.13	0.09	5.55	2.90	2.64	1.91
2	2029/30	96.4	9.16	2.02	7.14	0.09	7.04	2.90	4.13	2.42
5	2032/33	95.2	9.32	2.17	7.15	0.10	7.04	2.90	4.14	2.43
10	2037/38	93.3	9.60	2.45	7.15	0.12	7.02	2.90	4.12	2.42
12	2039/40	92.5	9.71	2.56	7.14	0.52	6.61	2.90	3.71	2.28
13	2040/41	92.1	9.76	2.63	7.14	0.40	5.23	0.00	5.23	-
15	2042/43	91.3	9.87	2.75	7.12	0.54	6.57	0.00	6.57	-
20	2047/48	89.3	10.16	3.10	7.06	0.54	6.51	0.00	6.51	-
25	2052/53	87.4	10.44	3.49	6.95	0.70	7.96	0.00	7.96	-

Year-13 CFADS reflects the inverter replacement; year-1 and year-25 CFADS reflect working-capital build-up and release. Full 25-year table in Appendix C.

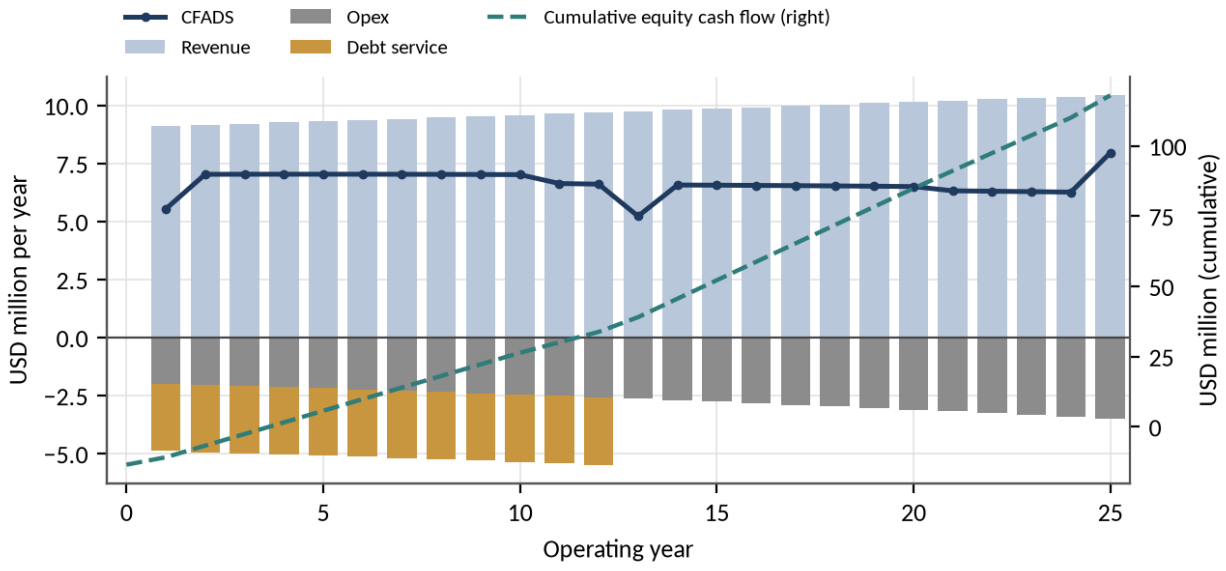


Figure 8.2 Annual cash flows and cumulative equity cash flow

Source: financial model.

8.8 Returns, payback and LCOE

Table 8.6 Investment returns, base case

Measure	Value	Comment
Project NPV at WACC	USD 9.93 million	Discounted at 15.36%
Project IRR	20.4%	Unlevered, after tax
Equity NPV at cost of equity	USD 2.80 million	Discounted at 24.35%
Equity IRR	28.8%	Levered, after tax
Project payback	4.8 years	Undiscounted, from COD
Equity payback	3.6 years	Undiscounted, from COD
LCOE at WACC	USD 0.0785/kWh	Nominal, pre-tax, all lifetime costs
LCOE at 8%	USD 0.0588/kWh	Compare IRENA 2025 global average USD 0.044/kWh [6]

The LCOE at an 8% discount rate is about 1.5 cents above IRENA's 2025 global average [6], which reflects the higher opex and modest yield of a fixed-tilt plant at 36 degrees north. At the project's own WACC the LCOE rises to 7.85 cents: in Syria the cost of capital, not equipment, drives the cost of solar power. Each percentage point off the WACC through guarantees or concessional finance lowers the cost of power for factories.

8.9 Debt service coverage

The minimum DSCR of 1.91x occurs in year 1, when the receivables balance is built up; from year 2 cover is about 2.4x (Figure 8.3). The loan life coverage ratio is 2.34x. Cover this high shows the debt is limited by gearing rather than cash flow. The study keeps it so because the headroom absorbs currency and collection shocks: the minimum DSCR stays above 1.30x until the tariff falls below USD 0.076/kWh.

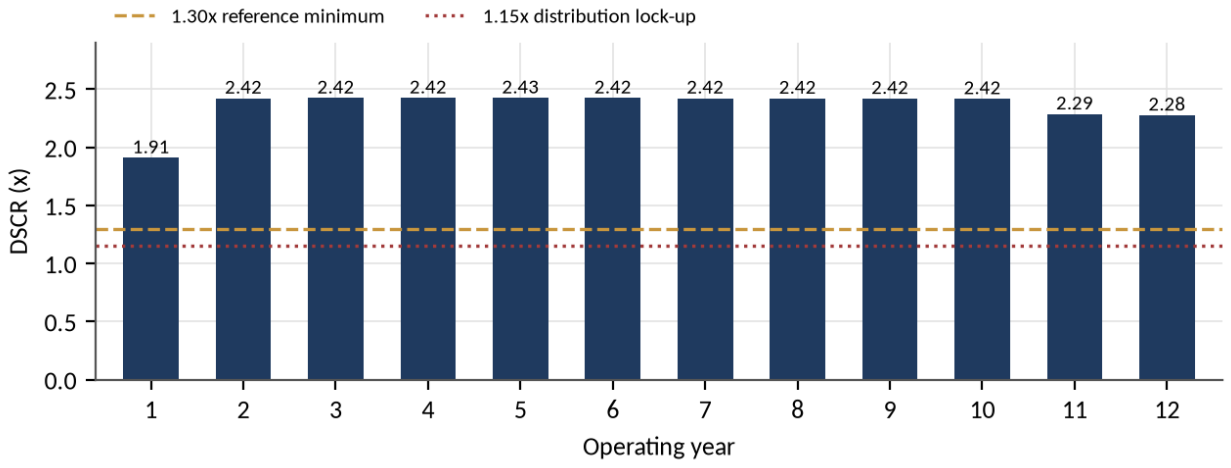


Figure 8.3 Debt service coverage ratio by year

Source: financial model.

8.10 Battery storage and tracker options

Table 8.7 Option economics

Indicator	PV only (base)	PV + 20 MW / 80 MWh BESS	PV with trackers
Capex (USD million)	32.2	45.4	36.6
Project NPV (USD million)	9.9	0.8	16.0
Project IRR	20.4%	15.7%	22.5%
Equity IRR	28.8%	19.4%	33.1%
Minimum DSCR	1.91x	1.37x	n/a

BESS: incremental NPV USD -9.1 million, incremental IRR 1.0%, break-even evening tariff USD 0.223/kWh.

The battery loses value because every kWh it shifts must first be bought from the plant at the daytime tariff, and it returns only 87% of that energy. At USD 0.16/kWh the spread is too thin to repay USD 13.2 million of capex. At an evening tariff of about USD 0.22/kWh, still 40% below diesel, the battery would break even, so it remains a credible second phase if evening offtake can be contracted.

9. Sensitivity, scenarios and Monte Carlo analysis

9.1 Sensitivity (tornado) analysis

Each driver was moved to a plausible adverse and favourable value with all others at base. The tariff, capex and currency conversion dominate; yield, opex and collections form a second tier; interest rate, availability, tax and degradation matter little.

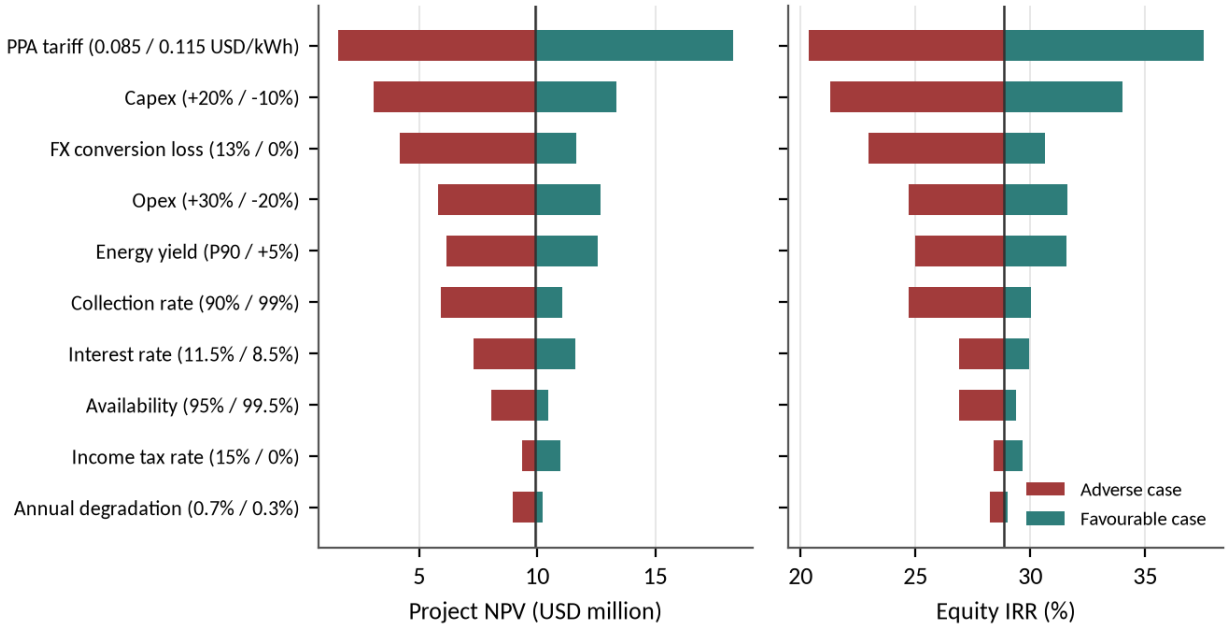


Figure 9.1 Tornado analysis: project NPV and equity IRR

Source: financial model, scripts/analysis.py.

Table 9.1 Sensitivity results

Driver (adverse / favourable)	NPV adverse	NPV favourable	Eq IRR adverse	Eq IRR favourable	Min DSCR adverse
PPA tariff (0.085 / 0.115 USD/kWh)	1.6	18.3	20.3%	37.5%	1.53
Capex (+20% / -10%)	3.1	13.4	21.3%	34.0%	1.58
FX conversion loss (13% / 0%)	4.2	11.7	23.0%	30.6%	1.65
Opex (+30% / -20%)	5.8	12.7	24.7%	31.6%	1.71
Energy yield (P90 / +5%)	6.2	12.6	25.0%	31.6%	1.74
Collection rate (90% / 99%)	5.9	11.1	24.7%	30.0%	1.73
Interest rate (11.5% / 8.5%)	7.3	11.6	26.9%	30.0%	1.74
Availability (95% / 99.5%)	8.1	10.5	26.9%	29.4%	1.83
Income tax rate (15% / 0%)	9.4	11.0	28.4%	29.7%	1.89
Annual degradation (0.7% / 0.3%)	9.0	10.2	28.2%	29.0%	1.91

NPV in USD million.

9.2 Break-even values

Table 9.2 Break-even values (other inputs at base case)

Variable	Break-even	Base case	Headroom
PPA tariff for project NPV = 0	USD 0.0822/kWh	USD 0.100/kWh	17.8%
PPA tariff for equity IRR = cost of equity	USD 0.0921/kWh	USD 0.100/kWh	7.9%
PPA tariff for minimum DSCR = 1.30x	USD 0.0760/kWh	USD 0.100/kWh	24.0%
Capex increase for project NPV = 0	+28.9%	USD 32.2 million	
Capex increase for equity IRR = cost of equity	+10.9%	USD 32.2 million	
Energy yield reduction for project NPV = 0	-18.9%	P50	
Collection rate for project NPV = 0	79.7%	97%	
FX conversion loss for project NPV = 0	20.3%	3%	

9.3 Scenarios

Table 9.3 Scenario results

Scenario	Capex	Project NPV	Project IRR	Equity IRR	LCOE	Min DSCR
Base case	32.2	9.9	20.4%	28.8%	0.0785	1.91
Upside	29.7	19.2	25.7%	40.2%	0.0689	2.41
Downside	37.1	-5.8	12.6%	13.9%	0.0975	1.23
Severe stress	40.3	-19.0	6.4%	2.5%	0.1122	0.81
Grid sale under Decision 154	32.2	-20.9	1.1%	-3.8%	0.0727	0.56
PV + 20 MW / 80 MWh BESS	45.4	0.8	15.7%	19.4%	0.1094	1.37

USD million; LCOE in USD/kWh. Definitions: Upside: Yield +4%, capex -8%, no FX loss, collection 99%, opex -10%; Downside: P90 yield (x0.928), capex +15%, FX loss 8%, collection 93%, opex +15%; Severe stress: P90 yield, capex +25%, FX loss 13%, collection 90%, opex +30%, debt 11.5%, no PPA indexation; Grid sale under Decision 154: All output sold to the state utility at 0.04 USD/kWh flat, no wheeling, full collection; PV + 20 MW / 80 MWh BESS: BESS charged from PV, evening energy at 0.16 USD/kWh.

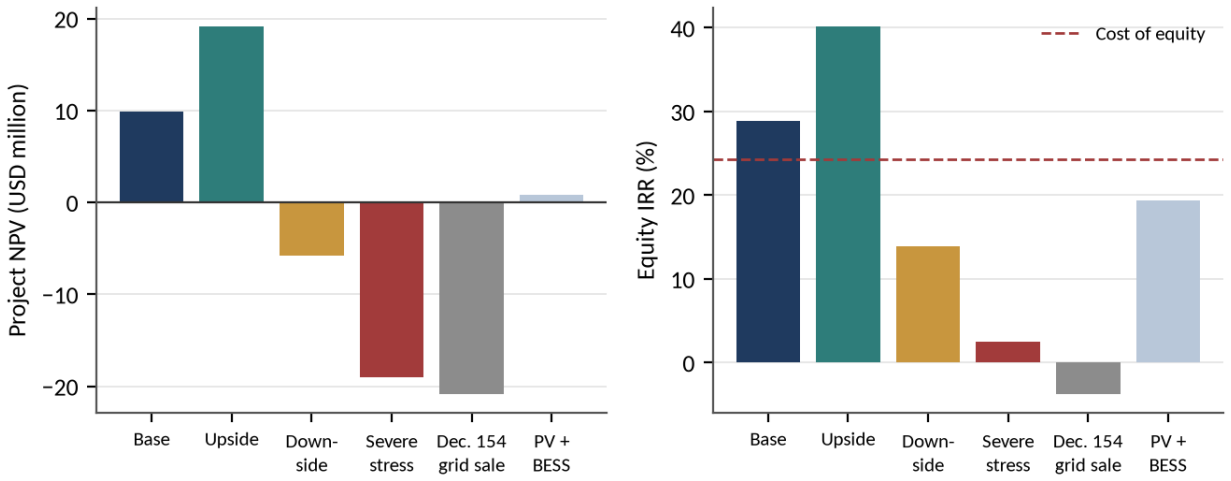


Figure 9.2 Scenario comparison

Source: financial model.

The downside case, which combines P90 yield, 15% higher capex, an 8% currency loss, 93% collections and 15% higher opex, gives an NPV of USD -5.8 million and a minimum DSCR of 1.23x: the debt is still served but equity falls well short of its cost. The severe stress case breaches debt service. Selling to the state under Decision 154 at USD 0.04/kWh destroys value even with full collections and no network charge, which confirms that the project is viable only as a direct industrial supplier.

9.4 Monte Carlo simulation

Table 9.4 Monte Carlo input distributions

Variable	Unmitigated case	With mitigants
Energy yield multiplier	Normal, mean 1.00, sd 0.056	Same
Capex multiplier	Triangular 0.92 / 1.00 / 1.25	Triangular 0.95 / 1.00 / 1.10 (fixed-price EPC)
Opex multiplier	Triangular 0.85 / 1.00 / 1.30	Same
FX conversion loss	Uniform 0-15%	Uniform 0-5% (convertibility undertaking, PRI)
Collection rate	Triangular 88% / 97% / 99%	Triangular 95% / 98% / 99% (guarantees)
Interest rate	Normal, mean 9.61%, sd 0.75%	Same
Annual degradation	Uniform 0.3-0.7%	Same
Availability	Triangular 95% / 98.5% / 99.5%	Same

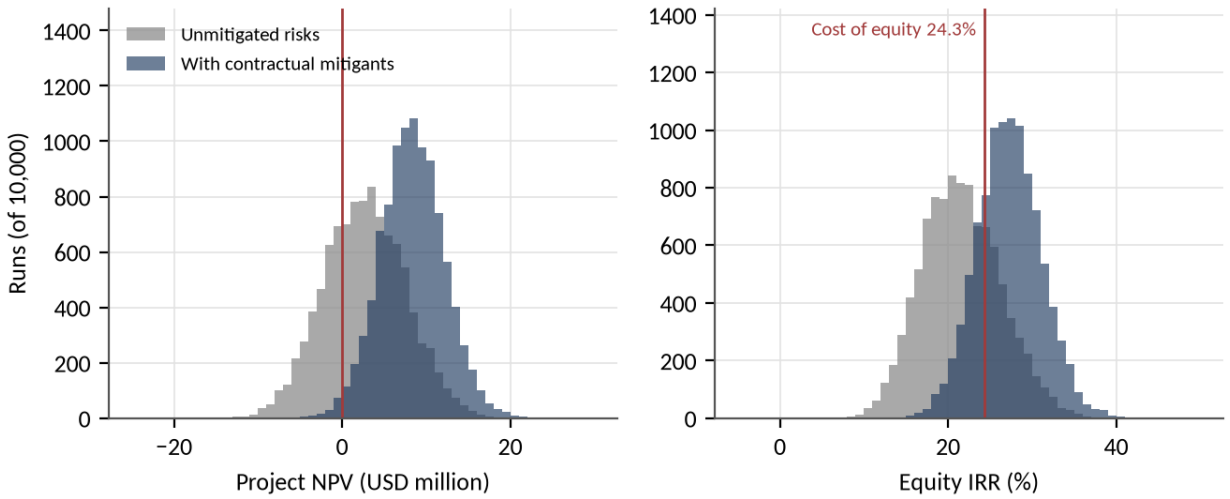


Figure 9.3 Monte Carlo distributions of project NPV and equity IRR (10,000 runs)

Source: scripts/analysis.py, seed 42.

Table 9.5 Monte Carlo results

Output	Unmitigated	With mitigants
Project NPV P10 / P50 / P90 (USD million)	-3.5 / 2.6 / 8.9	3.7 / 8.4 / 13.1
Probability of negative NPV	29.9%	1.3%
Equity IRR P10 / P50 / P90	15.7% / 21.3% / 27.8%	22.5% / 27.2% / 32.1%
Probability equity IRR below cost of equity	72.5%	22.2%
Minimum DSCR, 5th percentile	1.25x	1.56x
Probability minimum DSCR below 1.30x / 1.15x	8.4% / 1.4%	0.05% / 0.00%

The simulation's median NPV is below the base case because the input distributions are skewed to the downside: capex, opex and currency losses can rise further than they can fall. Unmitigated, the project is acceptable to lenders (minimum DSCR below 1.15x in only 1.4% of runs) but marginal for equity, which falls short of its 24.3% cost in 72.5% of runs. With mitigants that shortfall probability drops to 22.2%. This is the quantitative basis for the conditions in the recommendation.

10. Risk matrix and mitigants

Twelve principal risks were scored for likelihood and impact on a 1-5 scale before and after mitigation (Figure 10.1, Table 10.1). Currency convertibility, offtaker payment and the licensing route for direct sales are the risks that remain material after mitigation; they correspond to the conditions precedent in Section 1.

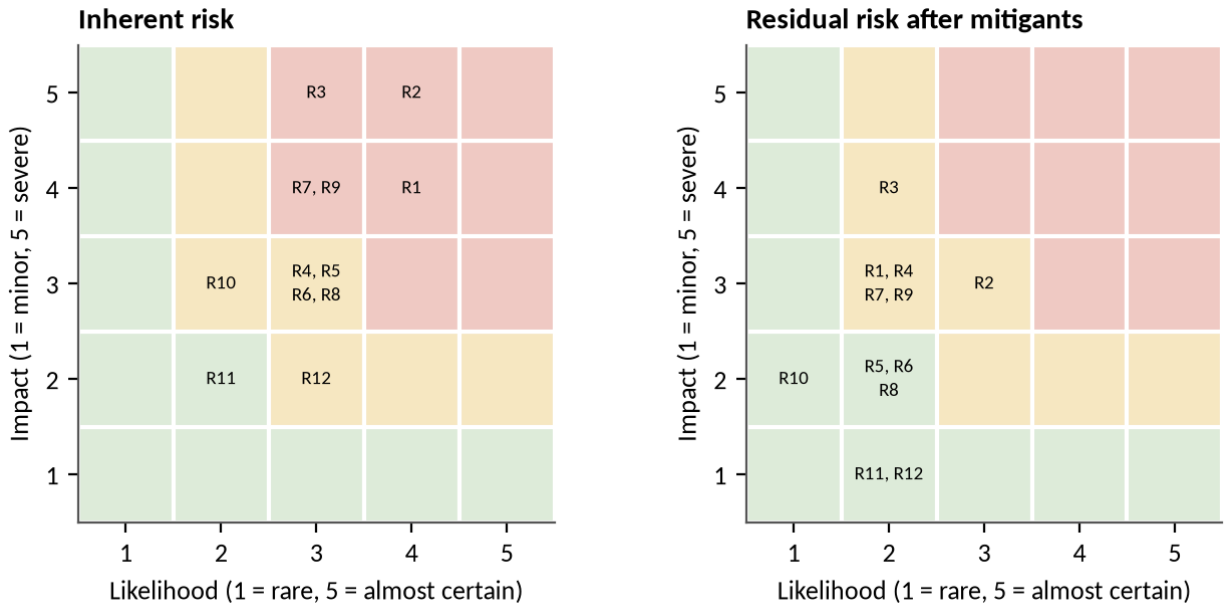


Figure 10.1 Risk heat map, inherent and residual

Source: risk register, data/risks.csv.

Table 10.1a Risk register (R1-R6)

ID	Category	Risk	L x I	Mitigants	Residual
R1	Offtake	Offtaker payment default or delay (industrial counterparties of mixed credit quality)	4x4	Portfolio of 15-30 offtakers with no single offtaker above 15% of volume; monthly prepayment or bank guarantee of 3 months' billing; right to suspend supply and re-allocate energy to other plants in the zone	2x3
R2	Currency	SYP depreciation or limited convertibility of SYP receipts into USD	4x5	USD-indexed tariff reset monthly at the CBS rate; convertibility and transfer undertaking under the Investment Law; political risk insurance; 6-month debt service reserve held offshore	3x3
R3	Regulatory	Direct sale to industrial users not licensed or restricted to the state single buyer (Decision 154 model)	3x5	Licence opinion and Ministry of Energy approval before financial close; structure under Law 41/2022 private-network provisions; fallback sale to Syrian Electricity Company priced by negotiated PPA	2x4
R4	Market	Grid supply to the zone improves and grid tariff in USD terms falls below the PPA price	3x3	Take-or-pay contracts for 10+ years; PPA discount to grid tariff; daytime energy priced below grid tariff at the official rate	2x3
R5	Construction	Capex overrun or delay (logistics through Latakia; security; customs)	3x3	Fixed-price date-certain EPC with liquidated damages; customs exemption under Investment Law confirmed before import; 10% contingency	2x2
R6	Technical	Lower energy yield than PVGIS estimate (dust soiling; heat; data uncertainty)	3x3	12-month on-site measurement campaign; P90 used for debt sizing; robotic dry cleaning; performance guarantee in O&M contract	2x2

Table 10.1b Risk register (R7-R12)

ID	Category	Risk	L x I	Mitigants	Residual
R7	Security	Theft or vandalism of cables and modules; security incidents	3x4	Perimeter fencing, CCTV and patrols; insurance including political violence cover; community employment programme	2x3
R8	Grid	Instability or outages on the zone network curtail plant output	3x3	Grid-forming inverter settings; network availability clause in PPA (deemed energy payments); interconnection study	2x2
R9	Financing	Lenders unable to close because of residual sanctions screening or correspondent banking limits	3x4	Sanctions and KYC due diligence on all counterparties; DFI-led lending group; sanctions-compliance covenants	2x3
R10	Environmental and social	Land-use conflict or inadequate ESIA delays permits	2x3	Site on industrial-zone land with no residents; ESIA under Ministerial Order 225/2008 aligned to IFC Performance Standards; grievance mechanism	1x2
R11	Tax	Draft income-tax law changes rates or incentives	2x2	Stability clause in investment licence; sensitivity shows limited effect on returns	2x1
R12	Operations	Spare parts and specialist skills unavailable locally	3x2	Two-year spare-parts stock; training of local technicians; O&M contract with response-time KPIs	2x1

11. Implementation timeline

Development would take about ten months to financial close in September 2027, followed by twelve months of procurement and construction to commercial operation in September 2028 (Figure 11.1). The critical path runs through offtaker contracting, the licence for direct sale and lender due diligence; procurement can start only after financial close because customs exemption and lender approval of the EPC contract are needed first.

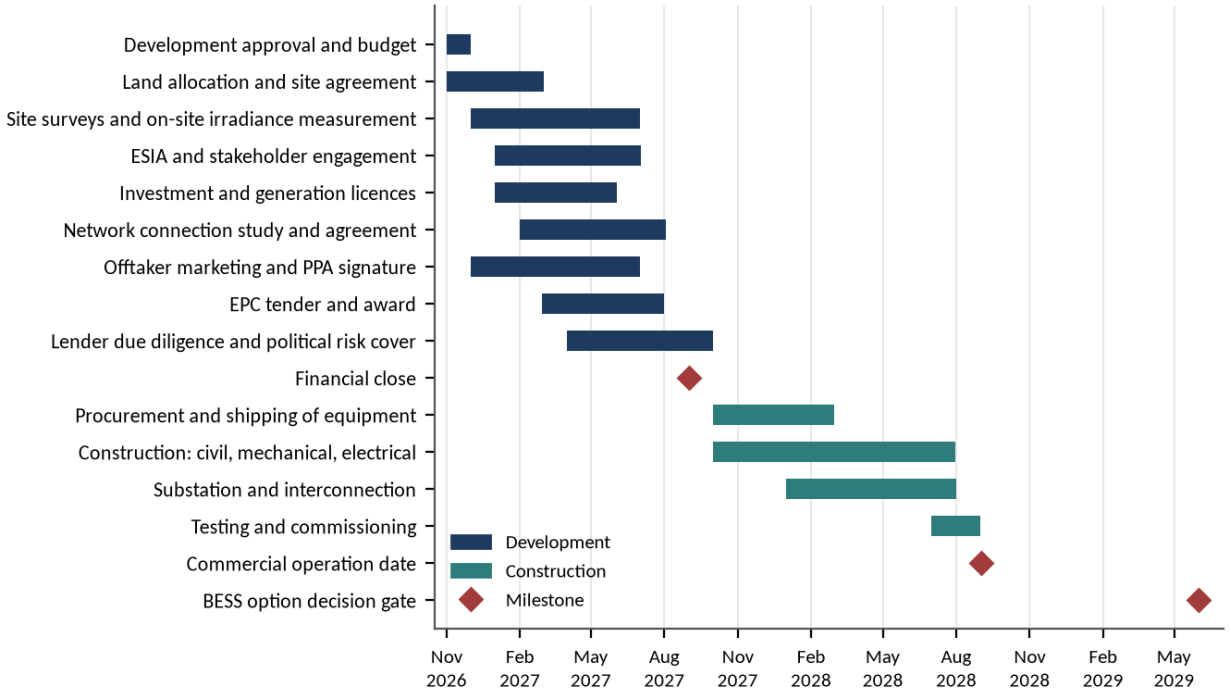


Figure 11.1 Implementation schedule

Source: Sponsor plan (assumption).

Table 11.1 Key milestones

Milestone	Target date	Gate condition
Development budget approved	November 2026	Board approval of USD 1.2 million
Site agreement signed	February 2027	Land allocation by industrial city
Anchor PPAs signed	June 2027	At least 80% of P50 output contracted
Licences issued	May-July 2027	Ministry of Energy confirmation of direct sale
Financial close	September 2027	All conditions precedent met
Commercial operation	September 2028	Performance tests passed
BESS decision gate	June 2029	Evening contracts at or above break-even tariff

12. Conclusion

A 60 MWp solar plant supplying factories at Sheikh Najjar is technically simple and commercially attractive in principle. The resource is good, at about 1,655 kWh/kWp from fixed-tilt modules, and equipment has rarely been cheaper. Daytime solar power at USD 0.10/kWh would undercut both the 17 SYP/kWh industrial grid tariff and diesel generation by a wide margin. On the base case the project returns 20.4% unlevered and 28.8% to equity, pays back in under five years and covers its debt about twice over.

What decides whether the project should be built is not engineering but contracts and institutions: whether direct sale to factories can be licensed, whether pound receipts can be converted to dollars reliably, and whether factories will sign and honour long-term payment commitments. The study therefore recommends proceeding to development with a clear set of gate conditions, deferring the battery until evening demand is contracted at about USD 0.22/kWh or more, and asking EPC bidders to price trackers alongside fixed tilt.

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Appendix A. Assumptions register

Every input used by the model is listed below with its basis. 'Sourced' values are taken directly from a dated source; 'Derived' values are calculated from sourced data; 'Assumption' values are analyst judgements and should be replaced with project-specific data during development.

Table A.1 Site, resource and design

Input	Value	Unit	Basis	Ref.	Note
lat	36.27	deg N	Assumption	[33]	Indicative site inside/adjacent to Sheikh Najjar Industrial City; existing PV plant reported at 36.2701N, 37.2569E
lon	37.3	deg E	Assumption	[33]	Indicative coordinates used for all resource queries
pvgis_yield	1,655	kWh/kWp/yr	Sourced	[1]	PVGIS 5.3, SARAH3 2005-2023, fixed 31 deg optimal tilt, 14% system loss
pvgis_sd_y	41	kWh/kWp/yr	Sourced	[1]	Year-to-year standard deviation reported by PVGIS
gsa_pvout	1,761	kWh/kWp/yr	Sourced	[4]	Global Solar Atlas PVOUT (Solargis model), cross-check only
model_uncert	0.05	fraction (1 sigma)	Assumption		Resource and modelling uncertainty used for P90 and Monte Carlo
mwp	60	MWp DC	Assumption		About a third of base-case estimated daytime zone demand (range 21-54%) so output is fully absorbed
mwac	50	MWac	Assumption		DC/AC ratio 1.20
availability	0.985	fraction	Assumption		Plant and offtaker-network availability; below the 99%+ typical in stable markets
deg_y1	0.01	fraction	Sourced	[9]	Jinko Tiger Neo warranty: max 1.0% in year 1
deg_annual	0.004	fraction/yr	Sourced	[9]	Jinko Tiger Neo warranty: 0.4%/yr years 2-30
land_ha_per_mwp	1.3	ha/MWp	Assumption	[32]	Existing Sheikh Najjar plant: 33 MW on 35 ha (1.06 ha/MW); 1.3 used for wider row spacing

Table A.2 Capital cost

Input	Value	Unit	Basis	Ref.	Note
c_modules	0.114	USD/Wp	Sourced	[5]	InfoLink 30 Sep 2026: TOPCon module average, non-China markets, before tax
c_logistics	0.012	USD/Wp	Assumption		Sea freight, insurance, Latakia port handling and inland haulage to Aleppo (about 10% of FOB)
c_inverters	0.03	USD/Wp	Assumption	[8]	PowerChina 2024 bids averaged 0.013 USD/W in China; export, delivery and spares margin applied
c_structure	0.065	USD/Wp	Assumption		Fixed-tilt galvanised steel mounting incl. piling
c_elec_bos	0.055	USD/Wp	Assumption		DC/AC cabling, combiner boxes, earthing, MV collection
c_substation_lump	4,000,000	USD	Assumption	[37]	Step-up substation and connection into the zone 20 kV network; zone already has 60 MW and 30 MW substations added in 2023
c_civil	0.035	USD/Wp	Assumption		Site preparation, roads, drainage, fencing
c_install	0.04	USD/Wp	Assumption		Mechanical and electrical installation labour
c_security_scada	0.012	USD/Wp	Assumption		CCTV, intrusion detection, SCADA, weather station
epc_margin	0.08	fraction of hard cost	Assumption		EPC overhead and margin
c_dev_lump	1,200,000	USD	Assumption		Feasibility, ESIA, legal, PPA negotiation, permits
owners_pct	0.015	fraction of EPC	Assumption		Owner's engineer and construction all-risk insurance
contingency	0.1	fraction of EPC	Assumption		Elevated for country risk and logistics

Table A.3 Operating cost

Input	Value	Unit	Basis	Ref.	Note
om_per_kwp	9	USD/kWp/yr	Assumption	[10]	O&M contract incl. cleaning; NREL ATB 2024 benchmark is 22 USD/kWac/yr all-in fixed O&M
insurance_pct	0.006	fraction of capex/yr	Assumption		Property and business interruption insurance
pri_pct_debt	0.008	fraction of initial debt/yr	Assumption		Political risk insurance premium on senior debt
security_fix	250,000	USD/yr	Assumption		Guarding and patrols
land_lease_per_ha	600	USD/ha/yr	Assumption		Lease of industrial-city land
admin_fix	300,000	USD/yr	Assumption		SPV management, audit, billing, metering
wheeling	0.005	USD/kWh	Assumption	[18]	Network use/metering charge for delivery over the zone distribution network
opex_esc	0.025	fraction/yr	Assumption		Opex escalation in USD
inverter_repl_yr	13	operating year	Assumption		Inverter replacement
inverter_repl_cost	0.025	USD/Wp	Assumption	[8]	Replacement cost in nominal USD

Table A.4 Revenue and market

Input	Value	Unit	Basis	Ref.	Note
ppa_day	0.1	USD/kWh	Assumption	[11]	USD-indexed PPA invoiced in SYP at the CBS official rate; 28% below the 17 SYP/kWh industrial grid tariff
ppa_esc	0.01	fraction/yr	Assumption		Annual PPA indexation
collection	0.97	fraction	Assumption		Share of invoices collected (bad debt 3%)
fx_haircut	0.03	fraction	Assumption	[15]	Expected loss converting SYP receipts to USD; official-parallel gap was 12.8% on 13 Oct 2026
receivable_days	60	days	Assumption		Debtor days for working capital
ppa_term	25	years	Assumption		Matches modelled life
grid_ind_syp	17	new SYP/kWh	Sourced	[11]	Decision 687: 1,700 old SYP/kWh for rationing-exempt industrial users; 100:1 redenomination
grid_heavy_syp	18	new SYP/kWh	Sourced	[11]	Smelting/rolling and high-consumption users: 1,800 old SYP/kWh
fx_official	122.5	new SYP/USD	Sourced	[15]	CBS official selling rate, 13 Oct 2026
fx_parallel	140.5	new SYP/USD	Sourced	[15]	Parallel market close, 13 Oct 2026
diesel_ind_syp	150	new SYP/L	Sourced	[13]	Second-grade diesel for productive sectors, announced 17 Sep 2026
diesel_std_syp	175	new SYP/L	Sourced	[12]	Standard diesel from 13 Sep 2026
genset_sfc	0.28	L/kWh	Assumption	[30]	Part-load factory gensets; MTU 20V4000 datasheet 202 g/kWh at full load (about 0.24 L/kWh)
genset_nonfuel	0.03	USD/kWh	Assumption		Genset maintenance, lubricants and capital recovery
dfi_ef_diesel	2.697	kg CO2/L	Derived	[31]	EPA 2025: 10.21 kg CO2/gal / 3.785 L/gal
fit_154_pv	0.04	USD/kWh	Sourced	[20]	Decision 154 (3 Jul 2025) feed-in tariff, standalone solar 2-10 MW
fit_154_bess	0.06	USD/kWh	Sourced	[20]	Decision 154 feed-in tariff, solar with 4-hour storage

Table A.5 Tax and financing

Input	Value	Unit	Basis	Ref.	Note
cit	0.1	fraction	Sourced	[26]	Draft unified income tax law: 10% for priority sectors (enactment not confirmed; treated as assumption-sensitive)
tax_relief	0.75	fraction	Sourced	[25]	Investment Law 18/2021 as amended by Decree 114/2025: up to 75% income-tax reduction for 10 years
tax_relief_years	10	years	Sourced	[25]	
dep_years	20	years	Assumption		Straight-line tax and book depreciation
gearing	0.6	fraction of funding	Assumption		Senior debt share; DFI-style lenders in frontier markets
ust7	0.0511	fraction	Sourced	[28]	7-year US Treasury par yield, 13 Oct 2026
ust10	0.0521	fraction	Sourced	[28]	10-year US Treasury par yield, 13 Oct 2026
sofr	0.0387	fraction	Sourced	[29]	SOFR, 9 Oct 2026 (reference only)
debt_margin	0.045	fraction	Assumption		Credit margin over the 7-year Treasury for a DFI-led loan with political risk cover
tenor	12	years after COD	Assumption		Annuity repayment
upfront_fee	0.02	fraction of debt	Assumption		Arrangement and commitment fees
idc_factor	0.5	years	Assumption		Average debt outstanding during a 12-month construction period
beta	0.85		Assumption		Levered equity beta for a contracted solar SPV
mature_erp	0.0423	fraction	Sourced	[27]	Damodaran mature-market ERP, Jan 2026
crp_syria	0.1554	fraction	Sourced	[27]	Damodaran Syria country risk premium (NR), Jan 2026
dscr_lockup	1.15	x	Assumption		Distribution lock-up covenant
dscr_target_min	1.3	x	Assumption		Minimum DSCR lenders would typically require for a merchant-exposed frontier project

Table A.6 Battery option and analysis settings

Input	Value	Unit	Basis	Ref.	Note
bess_mw	20	MW	Assumption		Option: 4-hour battery charged from PV
bess_mwh	80	MWh	Assumption		
bess_capex_kwh	165	USD/kWh	Assumption	[7]	BNEF 2025 global turnkey 117 USD/kWh; Ember all-in 125 USD/kWh ex China/US; +40 for Syria logistics and risk
bess_rte	0.87	fraction	Assumption	[7]	AC round-trip efficiency, LFP
bess_fade	0.02	fraction/yr	Assumption		Usable capacity fade
bess_om_pct	0.015	fraction of BESS capex/yr	Assumption		
bess_aug_yr	12	operating year	Assumption		Augmentation to restore capacity
bess_aug_cost	2,000,000	USD	Assumption		Nominal augmentation cost
ppa_evening	0.16	USD/kWh	Assumption		Evening (18:00-22:00) firm tariff; still well below diesel cost
life	25	years	Assumption		Operating life modelled; zero net terminal value (scrap offsets decommissioning)
cod_year	2028	calendar year	Assumption		COD September 2028 after financial close September 2027
mc_runs	10,000	runs	Assumption		Monte Carlo iterations, seed 42

Appendix B. Methodology and formulas

- Net present value: NPV = sum over t of CF(t) / (1 + r)^t, with year-0 capex undiscounted. Project cash flows are discounted at the WACC, equity cash flows at the cost of equity.
- Internal rate of return: the discount rate at which NPV equals zero.
- Payback: the year in which cumulative undiscounted cash flow turns positive, interpolated within the year.
- LCOE = [capex + sum of PV(opex + replacement capex)] / sum of PV(energy delivered), discounted at the WACC (and at 8% for benchmarking), nominal and before tax.
- CFADS = EBITDA - tax paid - increase in working capital - replacement capex. DSCR = CFADS / (interest + principal). LLCR = PV of CFADS over the loan life at the cost of debt / opening debt.
- Debt sizing: D = g x C / [1 - g x (fee + kd x 0.5)], which solves the funding loop between debt, fees and interest during construction without a circular reference.
- Cost of equity = risk-free rate + beta x mature-market ERP + country risk premium. WACC = E/V x Ke + D/V x Kd x (1 - t).
- P90 = P50 x (1 - 1.2816 x sigma), with sigma = sqrt(0.05^2 + (sd/P50)^2) = 0.0558.
- Tax: corporate tax on profit after depreciation (20-year straight line on total funding), interest and replacement capex, with losses carried forward.

Appendix C. Annual cash flows, base case

Table C.1 25-year cash flows (USD million unless stated)

Year	Period	GWh	Revenue	Opex	EBITDA	Interest	Principal	Tax	CFADS	Equity CF	DSCR
0	2027/28	-	-	-	-	-	-	-	-	-13.45	-
1	2028/29	96.8	9.11	1.98	7.13	1.94	0.97	0.09	5.55	2.64	1.91
2	2029/30	96.4	9.16	2.02	7.14	1.85	1.06	0.09	7.04	4.13	2.42
3	2030/31	96.0	9.21	2.07	7.14	1.74	1.16	0.09	7.04	4.14	2.42
4	2031/32	95.6	9.27	2.12	7.15	1.63	1.27	0.10	7.04	4.14	2.42
5	2032/33	95.2	9.32	2.17	7.15	1.51	1.39	0.10	7.04	4.14	2.43
6	2033/34	94.8	9.38	2.22	7.15	1.38	1.53	0.10	7.04	4.14	2.42
7	2034/35	94.4	9.43	2.28	7.15	1.23	1.67	0.11	7.04	4.13	2.42
8	2035/36	94.0	9.49	2.33	7.15	1.07	1.84	0.11	7.04	4.13	2.42
9	2036/37	93.6	9.54	2.39	7.15	0.89	2.01	0.11	7.03	4.13	2.42
10	2037/38	93.3	9.60	2.45	7.15	0.70	2.21	0.12	7.02	4.12	2.42
11	2038/39	92.9	9.65	2.50	7.15	0.49	2.42	0.50	6.64	3.74	2.29
12	2039/40	92.5	9.71	2.56	7.14	0.25	2.65	0.52	6.61	3.71	2.28
13	2040/41	92.1	9.76	2.63	7.14	0.00	0.00	0.40	5.23	5.23	-
14	2041/42	91.7	9.82	2.69	7.13	0.00	0.00	0.54	6.58	6.58	-
15	2042/43	91.3	9.87	2.75	7.12	0.00	0.00	0.54	6.57	6.57	-
16	2043/44	90.9	9.93	2.82	7.11	0.00	0.00	0.54	6.56	6.56	-
17	2044/45	90.5	9.99	2.89	7.10	0.00	0.00	0.54	6.55	6.55	-
18	2045/46	90.1	10.04	2.96	7.09	0.00	0.00	0.54	6.54	6.54	-
19	2046/47	89.7	10.10	3.03	7.07	0.00	0.00	0.54	6.52	6.52	-
20	2047/48	89.3	10.16	3.10	7.06	0.00	0.00	0.54	6.51	6.51	-
21	2048/49	89.0	10.21	3.17	7.04	0.00	0.00	0.70	6.33	6.33	-
22	2049/50	88.6	10.27	3.25	7.02	0.00	0.00	0.70	6.31	6.31	-
23	2050/51	88.2	10.33	3.33	7.00	0.00	0.00	0.70	6.29	6.29	-
24	2051/52	87.8	10.38	3.41	6.98	0.00	0.00	0.70	6.27	6.27	-
25	2052/53	87.4	10.44	3.49	6.95	0.00	0.00	0.70	7.96	7.96	-

Year 0 equity is the sponsor contribution at financial close.

Appendix D. Data gaps and limitations

- Industrial load: no published demand figure for Sheikh Najjar; the 93-233 MW range is an assumption based on facility counts. A metering survey is required.
- Existing solar capacity at the site: sources differ (33 MW contract in 2021; 42 MW operating plus 58 MW under construction per Global Energy Monitor).
- Network-use (wheeling) charges and licensing terms for direct industrial sales are not yet published; 0.5 US cents/kWh is assumed.
- Corporate income tax: the 2025 draft law (10% / 15%) is not confirmed as enacted.
- Grid emission factor for Syria is not published; CO2 savings are shown as an upper bound against diesel.
- Equipment prices other than modules (inverters, structures, BOS, civil works) are assumptions benchmarked to IRENA totals, not quotations.
- Diesel generator fuel efficiency in Sheikh Najjar factories is assumed (0.28 L/kWh); actual part-load efficiency varies.
- Exchange-rate path: the model treats currency risk as a conversion loss rather than forecasting the SYP.
- The resource assessment relies on satellite data; no ground measurements are available for the site.

Appendix E. Model verification

The workbook was recalculated in LibreOffice Calc (headless, full recalculation on load) and its outputs compared with the Python reference model. Result: ALL PASS; formula errors found: 0. The checks are listed below and are repeated live on the workbook's Checks sheet.

Table E.1 Integrity and reconciliation checks (part 1)

Check	Result
Sources equal uses: debt + equity - funding	PASS
Capex lines sum to PV capex	PASS
Debt fully repaid at end of tenor	PASS
Sum of principal equals debt	PASS
Equity cash flow = CFADS - debt service (all years)	PASS
Project NPV at project IRR equals zero	PASS
Equity NPV at equity IRR equals zero	PASS
Working capital released by final year	PASS
No negative tax paid	PASS
No negative energy	PASS
Gearing equals input	PASS
Minimum DSCR above lock-up covenant	PASS
Python: total capex	PASS
Python: senior debt	PASS
Python: year-1 energy (MWh)	PASS
Python: year-1 revenue	PASS

Table E.2 Integrity and reconciliation checks (part 2)

Check	Result
Python: year-1 EBITDA	PASS
Python: project NPV	PASS
Python: project IRR	PASS
Python: equity NPV	PASS
Python: equity IRR	PASS
Python: project payback	PASS
Python: equity payback	PASS
Python: LCOE at WACC	PASS
Python: minimum DSCR	PASS
Python: average DSCR	PASS
Python: LLCR	PASS
Python: WACC	PASS
Lever test: bess_on = 1 (Excel vs Python NPV and equity IRR)	PASS
Lever test: capex_mult = 1.2 (Excel vs Python NPV and equity IRR)	PASS
Lever test: yield_mult = 0.93 (Excel vs Python NPV and equity IRR)	PASS
Lever test: opex_mult = 1.3 (Excel vs Python NPV and equity IRR)	PASS